

Efficacy of remibrutinib in patients with CSU by demographics and baseline characteristics: Pooled analysis from REMIX-1 and REMIX-2 studies

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KEY FINDINGS & CONCLUSIONS

- This pooled analysis from the REMIX-1 and REMIX-2 studies showed that remibrutinib consistently improved CSU symptoms regardless of patient demographics, baseline characteristics and previous antihistamine treatment
- These outcomes indicate that remibrutinib is an effective oral treatment option across a broad range of patients with CSU who remain symptomatic despite treatment with second-generation H₁-AH
- Overall, remibrutinib also showed a favourable safety profile across the REMIX-1 and REMIX-2 studies

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INTRODUCTION

- Chronic spontaneous urticaria (CSU) is characterised by the spontaneous occurrence of itch, wheals (hives) and/or angioedema lasting for more than 6 weeks, without an identifiable trigger¹
- In the REMIX-1 and REMIX-2 studies, remibrutinib, an oral, highly selective Bruton's tyrosine kinase inhibitor, has shown superior efficacy vs placebo and a favourable safety profile when administered as an add-on medication in patients with CSU who remain symptomatic despite treatment with second-generation H₁-antihistamines (H₁-AH)²
- In the REMIX-1 and REMIX-2 studies, nearly half of the patients treated with remibrutinib achieved well-controlled disease (weekly Urticaria Activity Score [UAS7] ≤6) at week 12²

OBJECTIVE

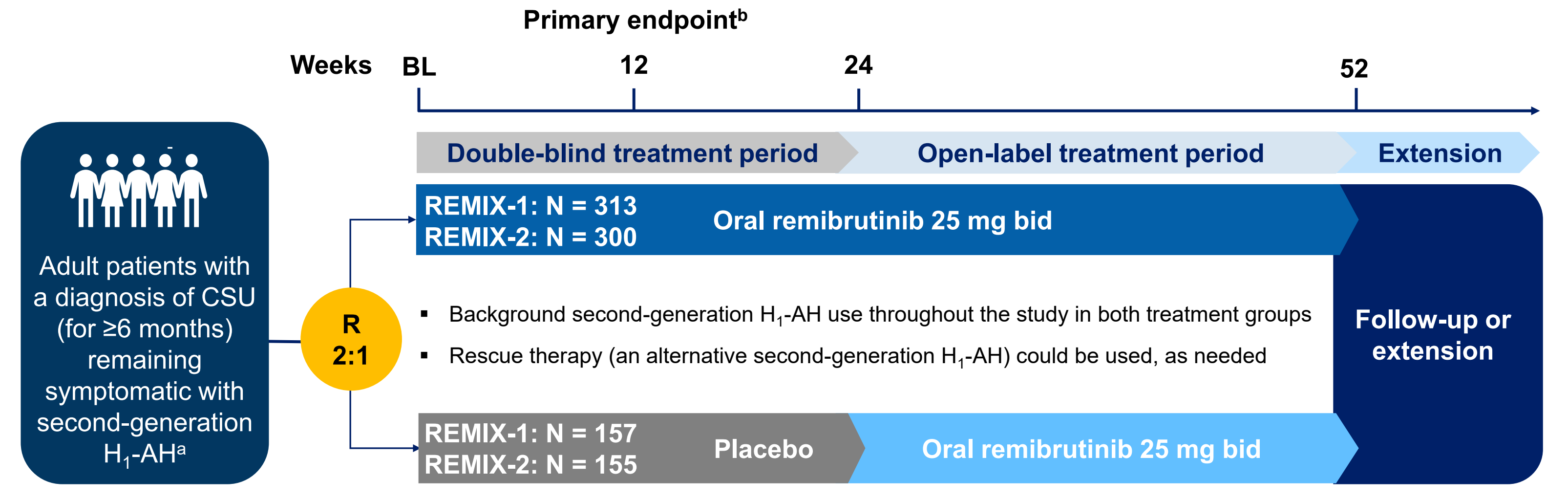
- This pooled analysis from the REMIX-1 and REMIX-2 studies aimed to assess the ability of remibrutinib to achieve well-controlled disease (UAS7 ≤6) in patients with moderate to severe CSU, categorised by baseline demographics and disease characteristics

METHODS

Study Design

- REMIX-1 and REMIX-2 were identical, multicentre, randomised, double-blind, placebo-controlled studies assessing the efficacy and safety of remibrutinib in adult patients with CSU who remained symptomatic despite treatment with second-generation H₁-AH
- Patients were randomised 2:1 to oral remibrutinib 25 mg twice daily (bid) or placebo over a 24-week double-blind period, followed by 28-week open-label treatment with oral remibrutinib 25 mg bid. At week 24, patients on placebo transitioned to remibrutinib (**Figure 1**)^{2,3}

Figure 1. Study design³



^aPresence of itch and hives for ≥6 consecutive weeks before screening despite the use of a second-generation H₁-AH; weekly Urticaria Activity Score of ≥16, weekly Itch Severity Score of ≥6 and weekly Hives Severity Score of ≥6 during the 7 days before randomisation (day 1).
^bThe primary endpoint was change in UAS7 from baseline to week 12.
bid, twice daily; BL, baseline; BMI, body mass index; CSU, chronic spontaneous urticaria; H₁-AH, H₁-antihistamines; N, number of patients; R, randomisation.

Study Assessments and Data Analysis

- The proportion of patients who achieved well-controlled disease (UAS7 ≤6) was analysed across subgroups, categorised by baseline age, gender, body mass index (BMI), CSU duration, previous experience of angioedema at baseline, disease severity and prior use of second-generation H₁-AH at a licensed dose and an up-dosing dose
- Data were analysed using a logistic regression model

RESULTS

- This analysis included 606 and 306 patients with CSU receiving remibrutinib and placebo, respectively, across the REMIX-1 and REMIX-2 studies
- In general, patient demographics and baseline characteristics were well balanced between the remibrutinib and placebo groups (**Table 1**)

Table 1. Patient demographics and baseline characteristics (pooled data)^a

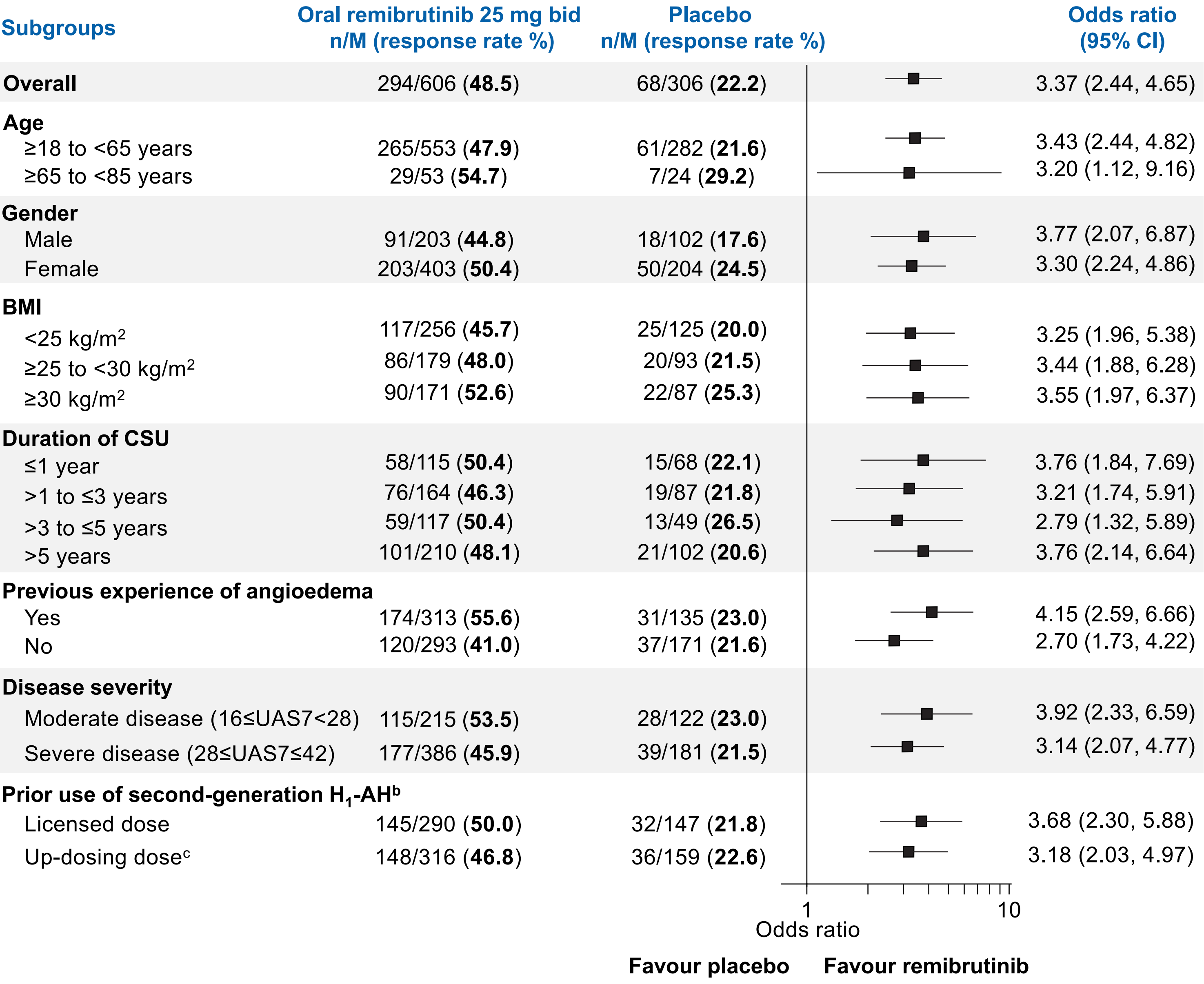
	Oral remibrutinib 25 mg bid (N = 606)	Placebo ^b (N = 306)
Age, years	43.3 ± 14.4	43.7 ± 14.1
Female, n (%)	403 (66.5)	204 (66.7)
BMI, kg/m ²	27.4 ± 6.5	27.7 ± 6.3 ^c
CSU duration, years	6.2 ± 8.6	5.3 ± 6.7
Previous experience of angioedema, n (%)	313 (51.7)	135 (44.1)
UAS7	30.6 ± 7.8	29.7 ± 7.6
Prior use of second-generation H ₁ -antihistamines, n (%)		
Licensed dose	290 (47.9)	147 (48.0)
Up-dosing dose	316 (52.1)	159 (52.0)

Data are presented as mean ± standard deviation unless specified otherwise.
^aFull analysis set. ^bPatients in the placebo group transitioned to remibrutinib at week 24. ^cn = 305.
bid, twice daily; BMI, body mass index; CSU, chronic spontaneous urticaria; N, total number of patients in the treatment group from the corresponding analysis set; n, number of patients who satisfy the designated criterion; UAS7, weekly Urticaria Activity Score.

Proportion of Patients Achieving UAS7 ≤6 Based on Subgroups

- Among patients categorised by baseline demographics, a higher number of patients achieved UAS7 ≤6 in the remibrutinib group vs placebo group (**Figure 1**)
 - Age groups of ≥18 to <65 years and ≥65 to <85 years at baseline: 47.9% vs 21.6% and 54.7% vs 29.2%
 - Male and female subgroups at baseline: 44.8% vs 17.6% and 50.4% vs 24.5%
 - BMI categories of <25 kg/m², ≥25 to <30 kg/m² and ≥30 kg/m² at baseline: 45.7% vs 20.0%, 48.0% vs 21.5% and 52.6% vs 25.3%
- Among patients categorised by disease characteristics, a higher number of patients achieved UAS7 ≤6 in the remibrutinib group vs placebo group (**Figure 2**)
 - Duration of CSU at baseline for ≤1 year: 50.4% vs 22.1%; >1 to ≤3 years: 46.3% vs 21.8%; >3 to ≤5 years: 50.4% vs 26.5% and >5 years: 48.1% vs 20.6%
 - Patients with previous experience of angioedema at baseline: 55.6% vs 23.0% and patients without previous experience of angioedema at baseline: 41.0% vs 21.6%
 - Patients with moderate disease (16≤UAS7<28) at baseline: 53.5% vs 23.0%, and patients with severe disease (28≤UAS7≤42) at baseline: 45.9% vs 21.5%
- Additionally, when patients were categorised by either the licensed dose or the up-dosing dose of second-generation H₁-AH at baseline, a higher proportion of patients treated with remibrutinib achieved UAS7 ≤6 (50.0% and 46.8%, respectively) vs placebo patients (21.8% and 22.6%, respectively; **Figure 2**)
- These findings suggest the favourable effect of remibrutinib in achieving well-controlled disease vs placebo across all subgroups evaluated

Figure 2. Forest plot for the proportion of patients achieving UAS7 ≤6 at week 12, categorised by baseline patient demographics and disease characteristics^a



Covariates included in the logistic regression model are treatment group, geographical region, prior exposure to anti-IgE biologics, baseline score, study, subgroup variable and interaction subgroup by treatment.
^aPooled full analysis set. ^bPrior dose of H₁-AH was defined based on the use of a second-generation H₁-AH at the licensed dose (local label-approved standard dose) or a second-generation H₁-AH at the up-dosing dose. ^cUp-dosing dose includes the use of second-generation H₁-AH for 2, 3 or 4 times a day.
bid, twice daily; BMI, body mass index; CI, confidence interval; CSU, chronic spontaneous urticaria; H₁-AH, H₁-antihistamines; IgE, immunoglobulin E; n, number of patients who responded; M, total number of patients in the treatment group with the response variable defined; UAS7, weekly Urticaria Activity Score.

Overall Safety Profile

- In the pooled safety analysis, remibrutinib showed favourable safety and tolerability across the REMIX-1 and REMIX-2 studies²
- The incidence of patients with at least 1 adverse event (AE) up to week 24 was comparable between the remibrutinib and placebo groups (64.9% and 64.7%, respectively)²
- Safety analysis by age and gender indicated no further risk of AEs or serious AEs (SAEs) in patients treated with remibrutinib vs placebo
 - The proportion of patients with at least 1 AE in the age groups of ≥18 to <65 years and ≥65 to <85 years was 63.3% vs 64.5% and 71.7% vs 66.7%, respectively
 - The proportion of male and female patients with at least 1 AE was 60.1% vs 60.8% and 67.2% vs 66.7%, respectively
- No deaths were reported, and discontinuation of the study treatment due to AEs was infrequent. No SAEs were considered related to the study treatment by the investigator²

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Disclosures

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